Gestalt

System card · Intent-driven workspace

Natural language → ranked needs → auction → morphing UI

Feature reference · prototype workspace (`/app`)

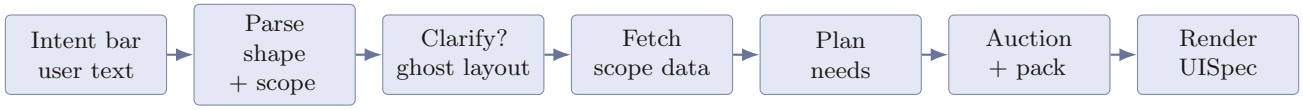
Abstract

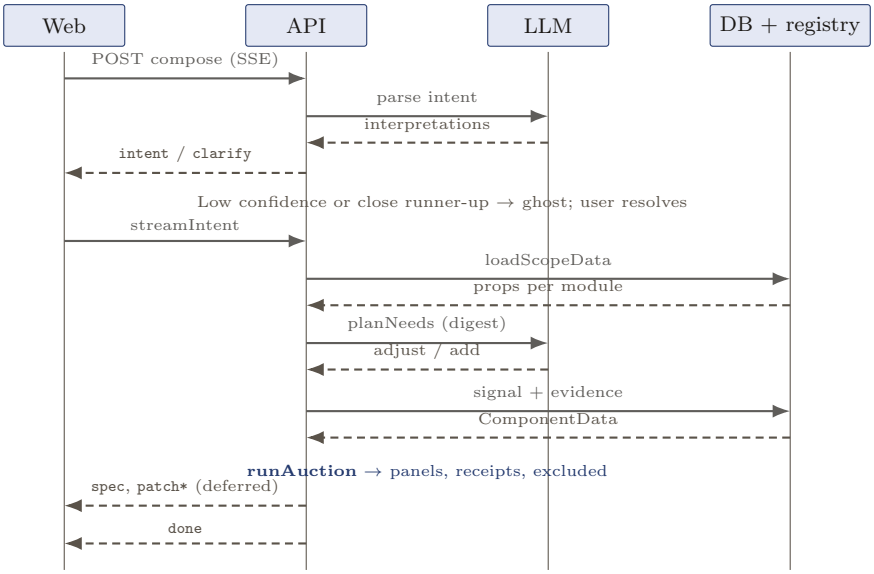
What we ship: a workspace that rearranges itself around the user's current job. We parse intent into a **shape** (task type) and **scope** (client, project, time window). We rank **needs** (information questions), run a **deterministic auction** over 17 panels, and render a **UISpec** the web client morphs in place. The model proposes meaning and priorities; **our code** owns layout, sizing, and exclusions.

1 Abbreviations & terms

Term	Meaning in Gestalt
LLM	Large language model. Used only for intent parsing, need-priority tuning, and prose in deferred panels—never for picking components or grid layout.
SSE	Server-Sent Events. `POST /session/:id/compose` streams stages (`parsing`, `intent`, `fetching`, ...) until `done`.
RAG	Retrieval-augmented generation. Optional chat sidecar; workspace composition does not depend on it.
API / Web	Express server (`apps/server`) and React client (`apps/web`).
DB	Postgres story data (clients, usage, tasks, meetings) loaded per scope; registry modules turn rows into panel **props**.
Registry	17 server modules (`apps/server/src/registry/*`)—one loader + signal + evidence per panel type.
UISpec	JSON bundle sent to the client: intent, needs, panels, receipts, exclusions, diff, degraded flag.
Intent	Resolved { `shape, scope, headline, summary, confidence` }—the job on screen.
Shape	Task type: `overview`, `prepare`, `diagnose`, or `triage` (see §**??**).
Scope	`clientId`, `projectId`, `timeRange` where time is `7d`, `30d`, or `90d`.
Need	A ranked question the UI must answer (e.g. "What changed in usage?"); not a React component id.
Manifest	Static metadata for a panel: which needs it answers (**fit**), allowed sizes, zone/order per shape.
Props	Typed payload for one panel instance (charts, lists, copy) after loaders run.
Digest	Compact fact summary for the scope; input to need planning and deferred LLM copy.
Signal (s)	Per-panel urgency in $[0, 1]$ from loaded facts; scales auction score by at most $\pm 30\%$.
Fit (f)	How well a panel answers a need ($[0, 1]$); from manifest `answers`.
Priority (p)	How much the current shape wants a need ($[0, 1]$); template + optional LLM adjust + pins.
Receipt	Explainability record for a placed panel: score, size, need, evidence strings, reason.
Why lens	UI mode showing receipts and grouped exclusions (not model-generated prose).
Intent bar	Primary NL input on the dashboard; triggers compose stream.
Ghost layout	Preview grid when intent is ambiguous—user picks an interpretation before full compose.
Fixture	Deterministic JSON fallback when the LLM is unavailable or for fixed multi-reading prompts.
Deferred panel	Layout chosen first; `talking_points` and `hypothesis_list` fill copy via LLM/fixture after `spec`.
Pin / Dismiss	Session prefs: pin forces a panel's need to priority 1.0; dismiss removes it until restored.
O / P / D / T	Shape columns in tables: **O**verview, **P**repare, **D**iagnose, **T**riage.
L / M / S	Panel sizes **L**arge, **M**edium, **S**mall (visual footprint on the grid).
u	Size **unit**: S = 1, M = 2, L = 3 for column packing budgets.
L6 / C8 / R6	Left column 6 u, center 8 u, right 6 u maximum per composition.
QBR	Quarterly business review (maps to `prepare` via keywords).
Jira	Atlassian issue tracker; live connector + `jira_tickets` panel when connected.
R / W	Read / write connector capabilities (role-gated).
eng	Engineering-tagged story tickets (developer role filter).
Δ%	Percent change in usage (or segment delta) vs baseline window.

2 Pipeline (end-to-end)





SSE order: `parsing` → `intent` → `fetching` → `planning` → `composing` → `spec` → `writing` → `patch*` → `done`.

3 Shapes & scope

Shape	Job we optimize for	Example headline
`overview`	Home, portfolio glance, sales chart, Jira board	Overview
`prepare`	Meeting / QBR prep	Acme meeting
`diagnose`	Explain a metric change	Acme usage
`triage`	Prioritize late or blocked work	Overdue work

Scope merge: omitted fields carry over from the current intent; explicit `null` clears. `prepare`/`diagnose` without a named client default to lowest-health client in scope.

3.1 Intent resolution (our precedence)

1. Multi-reading **fixture** when the prompt is fixed (deterministic clarify path).
2. Structured LLM call (temperature 0, 6 s timeout); unknown client/project ids dropped.
3. Single-reading fixture if the LLM fails.
4. Keyword parser (meeting→`prepare`, usage/drop→`diagnose`, overdue→`triage`, home→`overview`).

3.2 Clarify rule

We ask the user to choose when

$$c_1 < 0.6 \quad \text{or} \quad (c_1 - c_2) < 0.15,$$

where c_1, c_2 are top two interpretation confidences. UI shows ghost layouts per option, then compose. **Focused shortcuts:** sales-chart phrasing ⇒ `overview` + boost `sales_pulse`; Jira phrasing ⇒ `overview` + focus `jira_tickets` (center, size L).

4 Needs (questions, not widgets)

Each need is a question our UI must answer. Shape templates seed priorities; the planner may nudge from digest facts.

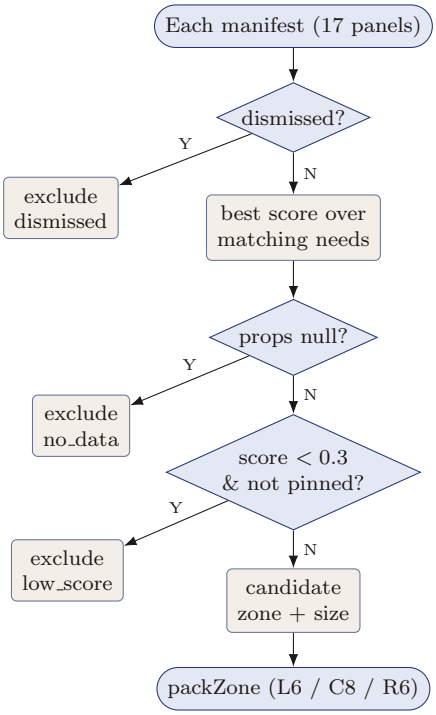


Enforced bounds: priority change ≤ 0.25; at most 2 added needs (priority cap 0.7); pins set primary need to 1.0; `overview` skips LLM planning.

Need type	O	P	D	T
portfolio_health / schedule	0.80 / 0.70	—	—	—
relationship / commitments / comms	—	0.90 / 0.85 / 0.70	0.40 / — / 0.45	0.30 / — / —
open_issues	0.50	0.80	0.60	0.40
talking_points	—	0.95	—	—
usage / segments / correlation / hypotheses	0.60 / — / — / —	0.75 / — / — / —	0.95 / 0.90 / 0.85 / 0.90	—
overdue / blockers / capacity	0.55 / — / —	—	—	0.95 / 0.90 / 0.80
sales_pulse / jira / actions	0.72 / 0.42 / 0.30	— / 0.55 / 0.60	— / 0.50 / 0.55	— / 0.62 / 0.70

Table 1: *
Template priorities by shape. "—" = need not in that shape's template.

5 Auction & layout



5.1 Score (auction bid)

$$\text{score} = p \times f \times (0.7 + 0.3\,s)$$

p = need priority, f = manifest fit, s = data signal (see abbreviations table).

Size from score. $\geq 0.8 \Rightarrow$ L; $\geq 0.5 \Rightarrow$ M; else S; then clamp to sizes the manifest allows.

Packing. Per zone: pinned first, then higher score, then stable id; shrink to fit budget; pin may evict unpinned; overflow $\Rightarrow$ `no_space`. Reading order from placement matrix (score breaks ties).

5.2 Default zones by shape (center = primary answer)

	Center	Left	Right
O	usage, overdue	portfolio, meetings	issues, comms, sales
P	talking points, usage	snapshot, last meeting	issues, Jira, actions
D	usage, segments, hypotheses	snapshot, releases	issues, comms
T	overdue, blockers, Jira	snapshot, workload	issues, actions

5.3 Signal (data urgency) — selected panels

Panel	How we compute s
portfolio_health	share of clients with health < 60
upcoming_meetings	1 if any meeting within 48 h, else 0.4
client_snapshot	$1 - \text{health}/100$
usage_trend / hypotheses	$\min(1, \lvert\Delta\%\rvert/40)$
overdue_queue	$\min(1, \text{overdueCount}/4)$
open_issues	$0.6 \cdot (\text{high/open}) + 0.4 \cdot$ recent-update share
jira_tickets	open + high-priority mix; 0 if disconnected
sales_pulse	deal volume and concentration
action_bar	constant 0.8

6 Workspace UI (what we render)

- **Headline** — `intent.headline` + `intent.summary` for the active job.
- **Three zones** — left (context), center (answer), right (next steps); ≥1100 px wide for three columns.
- **17 panels** — question titles; each bound to registry props + optional deferred copy.
- **Why lens** — auction receipts (score, size, evidence) and exclusion reasons.
- **Morphing** — `diff` drives add/remove animations; shared panels keep layout identity.

Home uses `overview` through the same auction (not a separate layout). **Pin / dismiss** recompose in place; **Home** clears prefs and recomposes overview.

7 Access roles (pre-auction filter)

Role	What changes in our pipeline
manager	Full data, sales panels, Jira read/write
developer	No sales/portfolio panels; eng-filtered tickets; Jira read
intern	Acme-only scope; hides sales, portfolio, workload, blockers, Jira, open issues

8 Example flow

Input: "Why did Acme usage drop?"

1. Intent → `diagnose`, client Acme, default `30d`; headline "Acme usage".
2. Template needs emphasize usage, segments, hypotheses, release correlation; planner may adjust ±0.25.
3. Strong usage Δ raises s; usage panel score ≥ 0.8 → center, size L.
4. Center: usage, segments, hypotheses; left: snapshot, releases; right: open issues.
5. Portfolio excluded (`not_relevant`)—no matching need on `diagnose`.
6. `hypothesis_list` streams copy after layout (`writing` / `patch`).

9 Feature guarantees

- **Separation of concerns:** LLM for language and priority hints; deterministic code for UI structure.
- **Explainability:** every visible panel has a receipt; every hidden panel has an exclusion reason.
- **Resilience:** fixtures and keyword intent when the model or DB is degraded (`degraded` flag on UISpec).
- **Optional chat:** RAG answers in the sidecar; Intent bar and auction path stand alone.

Implementation: `intent.ts`, `needs.ts`, `auction.ts`, `layout.ts`, `compose.ts`; shared manifests in `packages/shared`; compose SSE in `routes/session.ts`.